Integrating Optical Technologies for Enhanced Processing, Memory, and Bandwidth Scalability in Next-Generation Computing

A Bachelor's Thesis

Presented to

Department of Engineering Technology

State University of New York Polytechnic Institute

Utica, New York

In Partial Fulfillment

Of the Requirements for the

Bachelor of Science Degree

By

Ethan Khan (Processing), Owen Lawrence (Memory), and Devin DiTucci (Data Centers/Scalability)

Table of Contents

Abstract.....2
Introduction.....2
1 - Past Tech.....3
1.1 - The History of Computing.....3
1.2 - Early Memory Technologies..... 5
1.3 - How Data Centers Came to Be.....7
2 - Current Tech..... 9
2.1 - Processor Composition and Specifications.....10
2.2 - Modern Memory Innovations.....13
2.3 - Data Centers Today..... 14
2.4 - The End of Moore's Law.....16
3 - The Future of Optics in Computing.....19
3.1 - Optical Processing..... 19
3.2 - Photonic and Optoelectronic Memory.....23
3.3 - Optical Switching and Modulation in Data Centers.....26
Summary and Conclusions..... 29
References.....31

Abstract

In our present lives, personal computing has grown imperative to everyday life and work. Significant advancements have been made to the processing power and efficiency of computers nowadays. However, the growth of processing power has become somewhat stagnant due to reaching the physical limit of what can be innovated with silicon-based computing, with Moore's Law's observed trend of electrical transistor count reaching its finite limit of integration with ever-shrinking processor devices. To continue the significant growth of processing power in computing, it is necessary to integrate optical devices into consumer- and professional-grade computing. To this end, optical technologies will be utilized to improve upon modern personal computing in three ways: improvement on processor clock speeds, faster read-write capabilities and larger storage capacity for memory devices, and greater scalability and energy efficiency for data centers while promoting low latency and high data rates.

Introduction

Computers have become an integral part of human life, permeating themselves into all aspects and locations in our lives: they're in our hands to allow us to communicate with practically anyone on the globe at any point in time, they're in our homes entertaining and informing us with a seemingly limitless stream of news, information, and consumer content, and they're at our jobs as organizational and computational powerhouses to streamline our production. And just as we have grown and adapted to the ever-growing list of capabilities computers present themselves with in our lives, so too have they grown and adapted according to our needs. Computational power has increased exponentially to quicken every task, storage has grown in a similar manner to allow us to catalog and reference millions of documents and files at a time, and our data center industry has allowed every person and business to access the Internet at virtually any time with ever-increasing speeds.

However, the development of computing technologies has hit something of a snag; previously, the advancement of all of these technologies hinged on a manufacturer's ability to fit more and more transistors, the electromechanical switches that allow us to convert electrical power into a signal, onto each component. Gordon Moore, co-founder of Fairchild Semiconductor and Intel,

had coined a market tread that the number of these transistors that populated an integrated circuit would double every two years, and while we have shrunken these transistors to a point of being only microns large to keep up with this “law”, we have reached a point in the last decade or two where this is no longer a feasible option to continue the growth in computational capability we’ve been so used to since the 70s.

Not all is lost, though. We posit that processor, storage, and data center capability can continue its explosive growth as it has in the past, though through different means. Instead of committing purely to electromechanical means of computation and memory, integrating optical technologies into new and existing components can help lift development from the relative stagnancy it has found itself in. Specifically for each of these three aspects of computational advancement, we propose that the next generation of processing lies in the use of photonic integrated circuits and recurrent neural networks for data processing, photonic and optoelectronic devices for storage and memory, and in more widespread use of opto-electronic interfaces and optical switching technology, as well as the utilization of optical multiplexing in modulation devices for data center bandwidth capabilities.

1 - Past Tech

In order to fully understand how we intend to advance computer technology with optical and photonic components and methods, we must first look to the past as a point of reference for where we’ve come from and to integrate the lessons of the past into the endeavors of the future. To this end, the past iterations of computational, storage, and mass combinations of the two are listed below.

1.1 - The History of Computing

The history of optical processing dates back to the mid-20th century, when researchers first began exploring the use of light for computation due to its high speed and parallel processing capabilities. Opposed to a traditional electronic computing which is serial-based, optical computing’s parallel processing allows for higher processing of large datasets in real time. *Optical Computing: A 60-year Adventure (2010)* outlines the fundamentals of optical

information processing, stating the Fourier transform property of a lens is the basis of optical computing. When using coherent light, a lens performs in its back focal plane the Fourier transform of a 2D transparency located in its front focal plane. The exact Fourier transform with the amplitude and the phase is computed in an analog way by the lens. Following this, the development of holography from Dennis Gabor introduced new ways of storing and manipulating information using light interference patterns, setting the stage for analog optical computing. Most of these systems use lenses, lasers, and spatial light modulators (SLMs) to perform signal processing tasks.

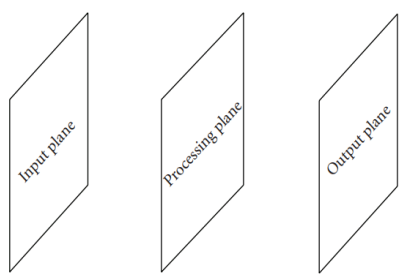


FIGURE 1: Architecture of an optical processor.

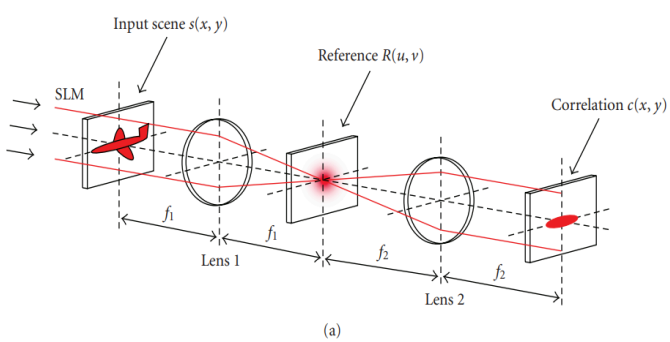


Figure 1.1: Architecture of an optical processor. 1.a Optical Setup. [1]

The era of electronic computing began in the mid-20th century with the development of machines like ENIAC and UNIVAC, which used vacuum tubes to perform basic arithmetic and logical operations. These systems marked a major change from mechanical computing, offering a much faster processing speed, though they were large, expensive, and consumed a great deal of power. The invention of the transistor in 1947 was a major turning point, enabling the construction of smaller, more reliable computers. The transistor led to the rapid miniaturization of components and the eventual creation of the integrated circuit in the late 1950s. By the 1970s

and 1980s, the microprocessor made computing for everyone. As transistors doubled, the observation by Gordon Moore, known as Moore's Law, predicted that computing power would increase exponentially. However, in current times, this prediction may be coming to an end due to the physical and thermal limits of traditional semiconductor devices [1].

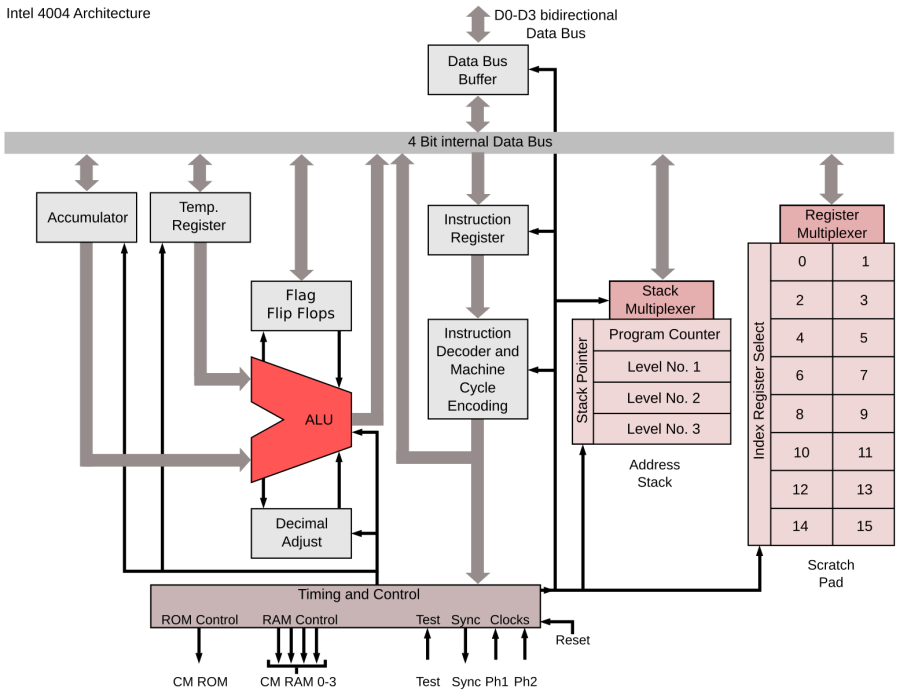


Figure 1.2: Intel 4004 architecture. [2]

1.2 - Early Memory Technologies

Evolution of Memory Architecture (2015) outlines how memory and computation are increasingly intertwined in response to new data, tasks, and technologies. Early and some recent computing systems relied on the classic von Neumann architecture, which physically separated memory and computation. While this approach has been effective, the growth of data and the changing nature of computational tasks have prompted innovations to address inefficiencies in the separation of memory and processing. The nature of these inefficiencies is also often described as the "Von Neumann Bottleneck" which represents the need for memory to be closer and closer to the main processor in order to be more effective [3].

In the past, the popular Symmetric Multiprocessing (SMP) paradigm allowed large pools of memory to be uniformly and coherently accessed by processors. However, processing-in-memory (PIM) systems, which processed memory inside of a memory module, faced limitations, such as restricted memory visible to processors on a single die, as well as limited capacity due to having to use up memory module space for processing. A promising alternative to PIM came to fruition involving relocating processing to intermediate layers between the processor and memory, such as the memory controller, this is most commonly in the form of a logic die where memory modules are stacked on top of the logic die allowing capacity of the total main memory to be significantly increased.

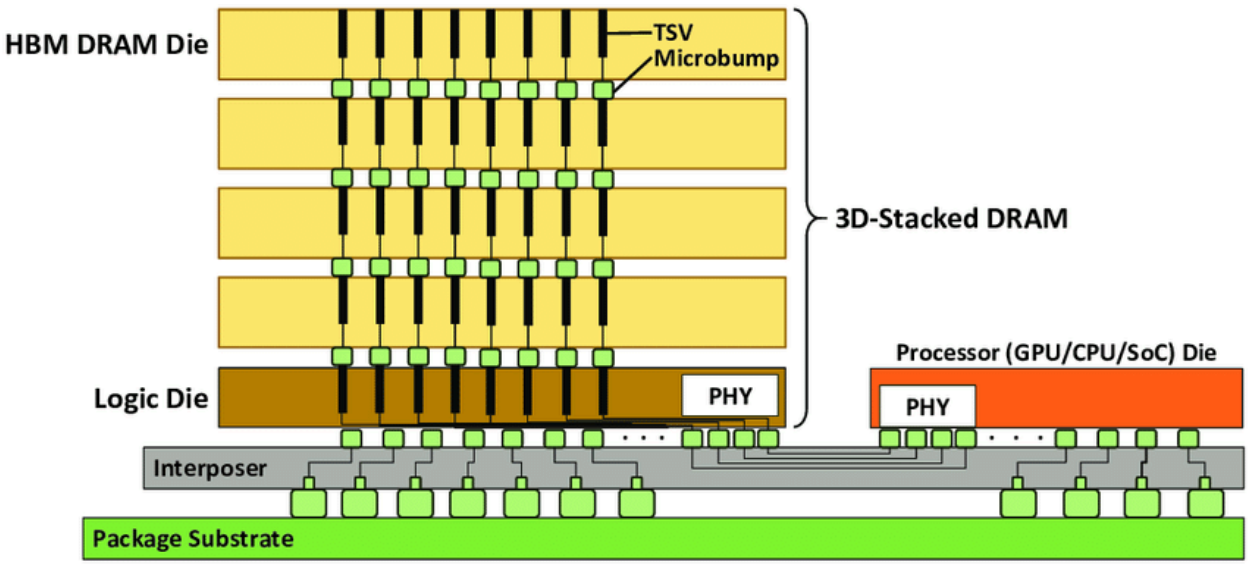


Figure 1.3: An example of 3D-stacked Memory. [4]

These memory controllers evolved to perform more advanced functions like atomic operations and integrating caches. With 3D-stacking technology, placing memory dies on top of logic dies significantly improves memory capacity, bandwidth, and energy efficiency. Micron's Hybrid Memory Cube (HMC) exemplifies such an architecture, using multiple DRAM layers and an integrated logic layer to manage data transformations, reducing bandwidth requirements and further improving performance.

During the time of DRAMs improvements to its storage early on, microprocessor innovations and clock speed improvements accelerated at an exponential rate, causing increased latency

between the processor and main memory, thus exacerbating the downfalls of the Von Neumann architecture. This was when Static Random Access Memory (SRAM) was introduced into caches extremely close to the processor. SRAM is different from DRAM in that SRAM does not need to be refreshed and is active as long as power is being supplied. SRAM can be accessed much faster and has much lower latency than DRAM, but is more expensive, requires more power and is less dense than DRAM, leading to a much smaller capacity of SRAM being located near the processor for the cache. The integration of SRAM into caches and logic controllers came about when the downfalls of DRAM in PIM became apparent. Early proposals of PIM and logic dies used DRAM for logic dies or for PIM but when it was realized that DRAM were only optimized for high densities at low-cost, the need for integration of SRAM in the logic die came to be.

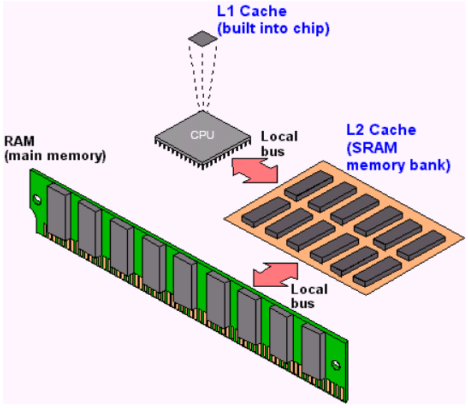


Figure 1.4: Shows the cache of SRAM close to the processor (further improvements to cache places an L1 cache built into the processor chip) [5]

These instances of innovation to memory at this stage shaped the way for the high-capacity RAM modules we see today, and integration of all the aforementioned technologies has led to increases in capacity of main memory by over 128 million-fold and equally impressive increases in RAM clock speeds.

1.3 - How Data Centers Came to Be

While the true beginning of mass information storage and distribution lies at the beginning of written history in the form of libraries and houses of learning, the first instance of a computerized data center came in the form of the Electronic Recording Machine - Accounting, or ERMA, in 1950. With the advent of bank checks as a form of payment in post-WWII America, banks and other financial institutions were drowning in a mountain of small paper slips with no efficient or organized way to sort and effect the changes these checks brought about on account holder's balances. As such, Bank of America went to the Stanford Research Institute,

who was then approached by both Texas Instruments and General Electric to build an automated check processing system to both sort and calculate monetary changes in their accounts [6].

The development of the first automated check-cashing machine was already a monumental step of progress in and of itself, but on top of this endeavor, Texas Instruments and General Electric both submitted proposals for processors that would replace the popularized vacuum tube system for newly-minted point contact transistors. This would allow the ERMA to process checks and account updates at speeds many times greater than a human bookkeeper could, the final GE ERMA reporting a then-whopping 250-kilocycle (250 kHz) clock speed, and increase technological shelf life, the transistorized boards never requiring as frequent of repairs or replacements as vacuum tubes had. This system also incorporated another groundbreaking processing technology, Magnetic Ink Character Recognition, or MICR. The account, routing, and check numbers for each individual check would be written at the bottom of the slip in a magnetic ink so that, when passed through the drum computer, the values printed upon the check would be automatically read and inputted for further processing down the line, as opposed to a secretary having to manually enter each value one at a time [7].

In order to record and store the data written herein, the ERMA would utilize magnetic tapes to work in collaboration with the MICR. These tapes, the most widely used of which being 10.5” tapes, hold 2400 feet of tape per reel, an equivalent of 140 MB of data storage [8]. With a system of eight tapes and transporters operating simultaneously at a read/write speed of 60 inches per second, the ERMA could process a gigabyte of data in the span of just over five minutes, unheard of for the time. Granted, this magnitude of processing power was never necessary as the ERMA would only ever need to process bytes or kilobytes, at most, of data at a time in terms of check numbers and values, but this incredible processing speed and mass storage of data would lead the ERMA to be considered one of the first data centers in existence, and a very powerful machine independently of any classification [7].

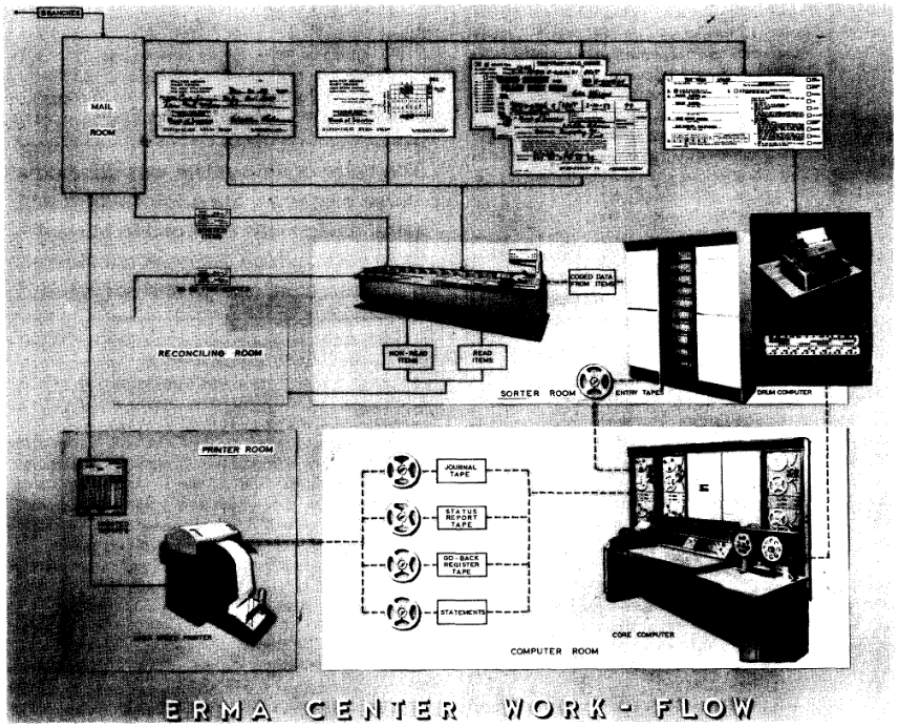


Figure 1.5: A process flow diagram for General Electric's final ERMA. Checks would first be sent to the sorter room, where their readability would be measured. If they could not be properly read, they would be sent to the reconciling room to either be re-processed or manually checked. If they could be read, the drum computer would sort the checks based on account number and intention (deposit or payment, for example), and then send this data to the core computer. This computer would put into effect the change that check would bring about on users' accounts then generate statements to be printed and stored [7].

2 - Current Tech

With these past developments as a solid stepping stone into further progress, computing technology would see an explosive growth in capability. Transistor technology would fully replace vacuum tube computing as the superior processing technology, and would continue to shrink in size while increasing speed and storage size for all devices. The sections below describe the fruits of those very labors, encapsulating what transistor computing is fully capable of and how far we've come from the latter half of the 1900s.

2.1 - Processor Composition and Specifications

Modern day computing is defined by the advanced, miniaturized, and efficient processors that power all systems. Central Processing Units (CPUs) have evolved to feature multiple cores and architectures capable of executing billions of instructions per second. Traditionally, CPUs are responsible for general-purpose tasks, but they have had significant improvements with hyper-threading, 3D stacking, and technology such as Scalable Coherency Fabric to enable faster inter-system communication. Intel's latest architectures, such as Arrow Lake, focus on hybrid-core designs that integrate high-performance (P) and energy-efficient (E) cores to optimize both multitasking and power consumption. However, some analysts suggest that as software becomes more multithreaded, this hybrid design may fall behind AMD's simpler yet highly effective approach. AMD's fourth-generation EPYC processors, have been regarded for having an innovative design, high performance, and energy efficiency. In the June 2024 article *AMD Next-Generation "Zen 4" Core and 4th Gen AMD EPYC Server CPUs* highlights a major advantage of having the Zen 4 core with AVX-512"Zen 4" adds support for AVX-512 instructions on a 256-bit data path. As a result, there is no change in peak per-cycle throughput using AVX-512 on "Zen 4." However, AVX-512 provides architectural advantages enabling performance improvements, including the following:

- Vector element masking support for nearly all instructions allows for reduced code vectorization overhead.
- 32, instead of 16, architectural registers for reduced register pressure.
- Code density benefits as one AVX-512 instruction can perform up to twice the amount of work of a 256-bit AVX2 instruction.
- Supporting 512-bit operations on a 256-bit data path is a power-efficient implementation with minimal area growth.

AMD currently offers the best available server and consumer level CPUs on the market.

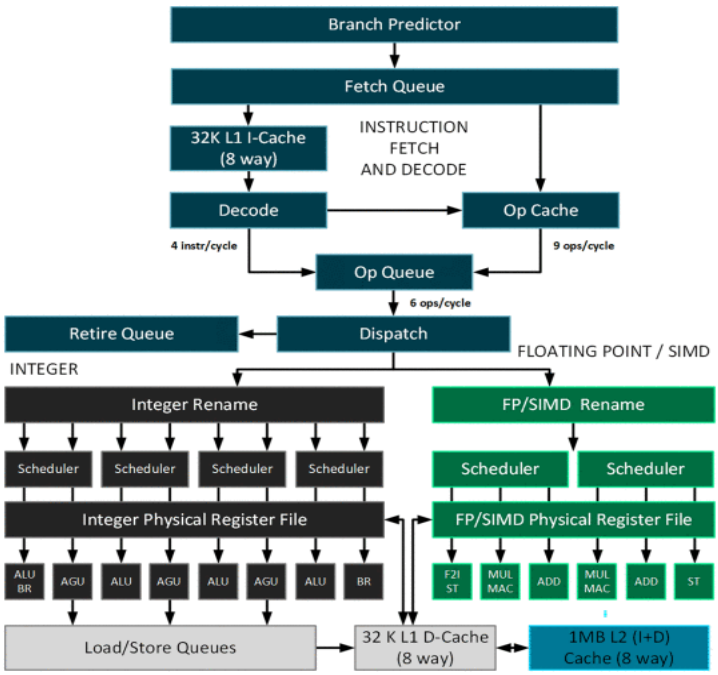


Figure 2.1: Zen 4 Microarchitecture block diagram. SIMD: Single Instruction, Multiple Data. [9]

Graphic Processing Units (GPUs) have become essential for parallel processing tasks, specifically in the areas of machine learning, scientific simulations, and real-time rendering. Today's processors are manufactured using advanced semiconductor fabrication technologies, with 5nm and even 3nm nodes in current production. Some of the new advancements include specialized units such as Neural Processing Units (NPUs) and Tensor Processing Units (TPUs). These are tailored for machine learning and AI tasks.Within these advancements are billions of transistors that fit on a single chip. NVIDIA has the best GPUs on the market with Hopper H100 being their flagship AI centered device. It is built on TSMC's 4N custom 4-nm process and contains over 80 billion transistors. The H100 development focused more on the architecture to improve upon its parallel programs of data locality and asynchronous execution.

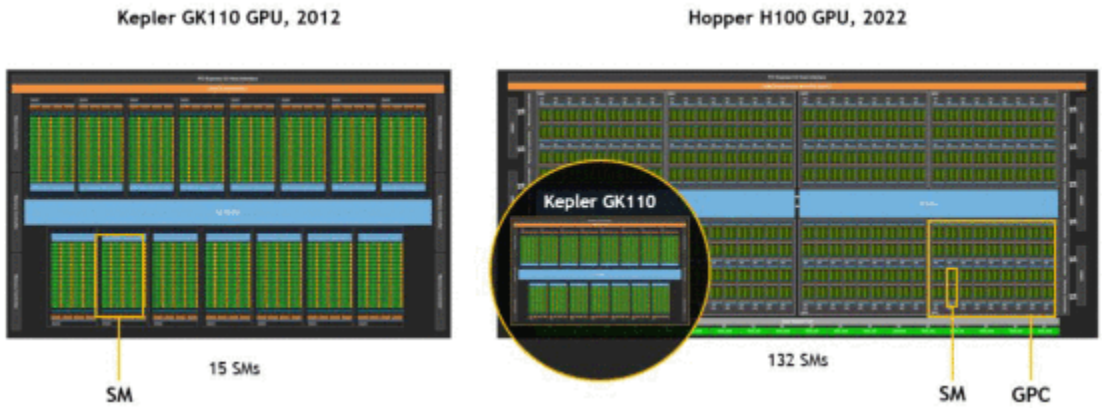


Figure 2.2: Orders of magnitude GPU scaling. SM: Streaming Multiprocessor. SM are the fundamental processing units that execute instructions in parallel, analogous to CPU cores. They contain CUDA cores and other resources such as registers and caches. [10]

Data locality is focused around moving and keeping program data as close to the execution units as possible. Asynchronous execution allows a program to continue executing while waiting for a long-running task to finish, rather than waiting for the task to complete before proceeding. Before Hopper, the architectures were not designed around the idea of AI models. The article *NVIDIA Hopper H100 GPU: Scaling Performance* highlights these hardware improvements. At the heart of the H100 SM is the new fourth-generation Tensor Core. The new Tensor Core has double the throughput per SM, clock-for-clock, for all data formats compared to A100. TF32, FP16, BFLOAT16, and INT8 have increased to 1024, 2048, 2048, and 4096 MACs/clock/SM, respectively. H100, like A100, supports sparse tensor arithmetic enabling an additional 2× throughput when one of the operands is sparse. To keep these Tensor Cores fed and to keep GPU power in check, H100's Tensor Cores have also improved operand delivery efficiency by 30%. All of these showcase NVIDIA's active pursuit and engineering of the next generation of GPUs for the demands of Future Artificial Intelligence [10].

The demand for computing power continues to surge with new emerging technologies such as autonomous vehicles, cloud computing, quantum computing, advanced signal processing, and (LLMs) Large Language Models. Traditional scaling has its limitations with problems such as thermals and signal delays.

2.2 - Modern Memory Innovations

In current day memory technology, 3D-stacked high-bandwidth DRAM modules (HBMs) still dominate the mainstream memory market today. While new-age RAM technology like Spin-Transfer Torque RAM, a non-volatile memory technology that uses magnetic tunnel junctions (MTJs) to store data, or phase-change memory (PCMs) which is a non-volatile (meaning it does not need to be refreshed like DRAM) memory technology that stores data by changing the physical state of a material between amorphous and crystalline states using electrical current pulses to achieve these transitions are presently being researched, there are economic and physical challenges to integrating them into mainstream technologies widespread.

In a presentation from the August 2015 *Hot Chips GPU Session* conference, details about the new HBM technology being implemented and how it is superior to previous graphics double data rate (GDDR) memory packages are discussed. This memory is specifically designed to be stacked and on an interposer, with each HBM DRAM die having through-silicon vias (TSV) interconnects between each other. These interconnects also connect the SoC/CPU to the interposer. This HBM-stacking technology allows the module to have a vastly higher bandwidth than a GDDR memory package while also operating at a lower clock speed and lower voltage. The HBMs also occupy significantly less space on an interposer, meaning that there is more space for a higher capacity of memory. In main memory, HBMs are not utilized due to higher cost and Dual-Inline Memory Modules (DIMMs, which are the most common packages for main memory modules) uses Double-Data Rate (DDR) processing protocol, which requires data to be processed on both the rising and falling edges of the clock signal. It is also why DIMMs are sold in pairs of 2, to utilize dual-channel processing for DDR-based memory. However, even with this limitation, innovations to the capacity and transfer speed of DDR-based DIMMs have still been made. These generational leaps in DDR technology are signified with numbers, with the latest being DDR5 which has transfer speeds going up to 8800 MegaTransfers per second. Capacity has also been increased dramatically, with more heat dissipation technology allowing for more modules to cover the DIMM [11].

Figure 2.3: A very high-density, tall-form factor DDR5 DIMM from Micron that holds 256GB of DRAM modules. [12]

3D-stacked High-Bandwidth Memory continues to set the standard for high-performance memory in specialized applications, offering significant advantages in bandwidth, power efficiency, and form factor through innovative design elements like through-silicon vias and interposers. Despite HBMs not being suited for widespread use in main memory due to their higher cost and architectural incompatibilities with existing DIMM infrastructure, traditional DDR-based memory continues to make improvements with data transfer rates and module capacities. Yet, due to the physical limits of what can fit on DIMMs as well as data transfer speeds for silicon memory reaching a ceiling, the innovations for purely silicon-based memory have been coming to a standstill. This is precisely where PCMs and optical based memory controllers can solve this issue and innovations to the processing and data transfer speeds of memory can continue to be made.

2.3 - Data Centers Today

As data processing speed and storage size have both increased explosively, the demand for greater quantities of data all at once has increased as well. Since the days of room-sized single computing devices, data center technology has evolved, while still remaining room- or even building-sized structures, to shrink each of the necessary devices from monolithic-sized

consoles to briefcase-sized single devices stored vertically in racks. Each of these single rack devices would be comprised of a central processing unit (CPU), multiple volatile memory modules alongside more permanent data storage units, input/output (I/O) cards for receiving and transmitting data across multiple channels, and all the power and cooling apparati necessary in order to operate each device at full capacity without overheating it in close quarters with racks and human-sized towers of similar units.

These data centers act as a consumer's full connection to the Internet, facilitating access to the World Wide Web on behalf of the consumer, at the consumer's request. From applications such as web-based email, streaming, social media, and archival data access, high-speed and high-volume data connections are a necessity for unadulterated Internet access. *The Datacenter as a Computer,* a consolidation of the standardization, construction, and resource cost that go into the warehouse-scale computers that go into forming data centers, specifically notes the Google Jupiter as an example of a modern, high-end data center configuration for powering data processing. Utilizing silicon dioxide metal oxide semiconductor field-effect transistors (MOSFETSs) and electrical switches, each Google Jupiter tower holds sixteen 40 Gbps ports (combination of 256 10-Gbps ports), allowing for fabric and host speeds of 10 Gbps or 40 Gbps, depending on the port connection. A single tower within each data center consumes a total of 10kW per hour, with each individual switching device consuming 250 W in the same stretch of time [13].

Dell has also listed three different switches used in data center configurations that have even exceeded the Google Jupiter in terms of data rates and number of ports. Each model of switch, the Z9864F-ON, Z9664F-ON, and Z9342F-ON models, list configurations in which they are all capable of utilizing 400-Gb ports (128, 64, and 32 ports, respectively), but the first of these models is capable, at maximum, of supporting 64 800-Gb ports. However, this throughput comes at a cost. The standard operating power cost to operate each server rack runs upwards of 500 W, all the way up to 1500 W, and when they max out under heavy load, they can consume almost twice as much power. These power costs also do not account for the additional devices attached to these switches to help facilitate mass data transit, such as multiplexers and demultiplexers, nor do they account for the power consumption of their surrounding elements. Monitoring stations,

fire suppression systems, external cooling, and all the power failsafes in place lend towards extreme power consumption in data centers, with estimated values in the 10s to even hundreds of kilowatts for standard operation [14].

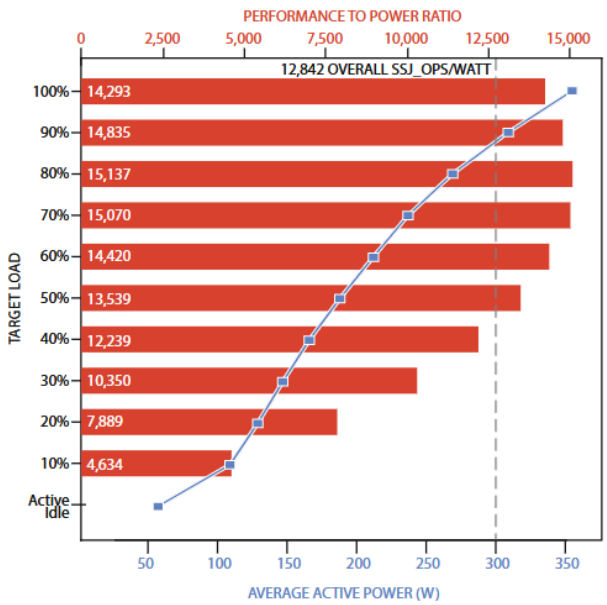


Figure 2.4: A measure of the power consumed per server device under different computational workloads based on a 28-core Intel Xeon with 192GB DRAM and an M.2 SATA SSD. [15]

Alongside these concerns of power consumption, bandwidth demands have continued to increase with the improvement and easier consumer access to streaming services, the constant increasing of file sizes, and the growth in the amount of people who have access to the Internet, to a point where current server technologies are struggling to keep up. In order for the industry of mass data transfer and storage to continue to be economically viable, power consumption has to drop and bandwidth capabilities must continue to increase with the general populace.

2.4 - The End of Moore's Law

This paper opened with an introduction to Moore's Law, the economical trend coined by Fairchild Semiconductor and Intel co-founder Gordon Moore, that the number of transistors in an integrated circuit would double every two years. As stated then, this "law" has been foundational

in benchmarking the progress made in the capabilities of personal and large-scale computing in all its aspects, both exponentially increasing processing speed and storage capacity while shrinking the sizes of every singular computing device from taking up an entire room or three to fitting snugly on our desks or in our hands. However, all good things must come to an end eventually, and the massive rate of advancement Moore's Law predicted for us is no exception.

Firstly, we have reached a physical size limitation in the amount we can shrink the transistors in order to integrate more of them into our devices. While these components could still potentially continue to miniaturize, the issue we as a collective have run into is that the insulator thickness has reached a point where if it shrinks any more than it already has, there would be devastating losses due to leakage current in both positions, either a decrease in switching speed in the "on" position or an increase in passive power consumption in the "off" position. Integrated circuit manufacturers and research engineers have attempted to adapt to this failing with new combinations and constructions of MOS materials and differentiating the method of switching for transistors, and while these adaptations have yielded positive results in continuing to miniaturize devices a little bit more, we are still running into the same issue as before. Not to mention, the time it's taken to research, test, and release these new adaptive methods has eaten into the time it takes to keep in the range of Moore's Law in terms of how many transistors we can continue to integrate [16].

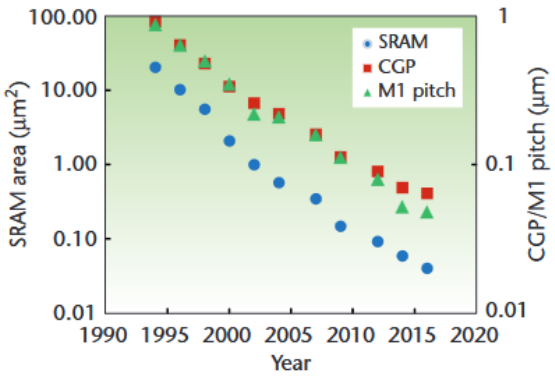


Figure 2.5: A measure of the rate of downsizing of three key variables in integration density over time. The blue dots represent the size and density of static random access memory (SRAM) in an integrated circuit device, the red squares represent contacted gate pitch (CGP), which is the minimum spacing between transistors, and the green triangles represent M1 pitch, the minimum wire-to-wire spacing in the first wiring layer. While each component and

section of spacing has continued to downsize over the measure of ~30 years, it is plain to see that they have not been able to keep a consistent trend of shrinkage, thus highlighting the issues stated above. [16]

The second major issue in the upkeep of Moore's Law as a continuation of progress in computing devices is the slowing decrease in energy efficiency, specifically in keeping the switching energies in individual transistors down. Similar to the downsizing issue, further research and development by companies such as IBM and Intel have continued decreasing the energy dissipation per state switch, however, these decreases have not been able to keep up. A study done in 1988 by German-American physicist Rolf Landauer extrapolated the trend of increasing energy efficiency as processing technology evolved from vacuum tubes to point-contact and bipolar transistors, then to modern field-effect transistors, following a very similar average trend to Moore's Law and being labeled as a kT constant. Modern companies and technologies have attempted to keep up with this trend, but it is increasingly clear that, similar to the size of the transistors themselves, we have reached a point where it has become very difficult to continue making transistors more energy efficient.

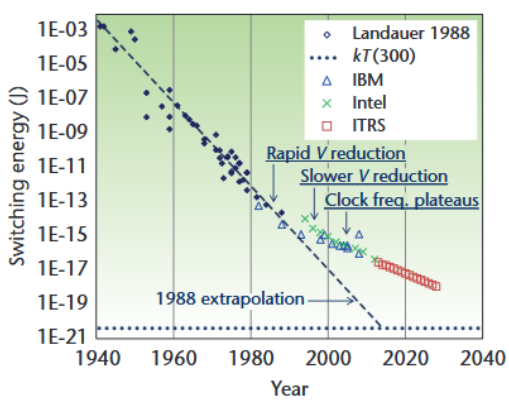


Figure 2.6: A visual representation of the aforementioned trend of switching energy efficiencies of field-effect transistors. The black diamonds represent the results of Landauer's study, which form the constant kT dotted line shown here. While companies such as IBM (blue triangles) and Intel (green Xs), and even a study completed by the 2013 International Technology Roadmap for Semiconductors (ITRS, red squares) have attempted to continue along Landauer's trend, the decrease of voltage reduction by transistors' state switching, along with the subsequent increases in clock frequencies brought on by this increased efficiency, have both plateaued. [16]

3 - The Future of Optics in Computing

Both of these issues with the effects of Moore's Law, along with the compiling list of other difficulties described in the above sections, seem to call for an alternative to purely-electrical technologies powering computing devices. Luckily, there have been very recent developments and studies done of integrating optical technologies into different aspects of computers and data centers, showing a continued increase in the capability of their respective components and devices while cutting down on resource consumption. Below are examples and descriptions of these technologies, with the hope that further work, as needed, will see commercial product integration for the betterment of the future of computers.

3.1 - Optical Processing

As we approach the limitations of electronic computing in both its physical and practical limits, optical computing showcases a promising path for sustaining and surpassing current performance benchmarks. Unlike traditional electronics, which use electrons, optics use photons, which travel at the speed of light and do not generate heat via resistance, making them ideal for high-speed, low-power data transmission. Future optical computing systems are expected to leverage these options to achieve unparalleled bandwidth, faster data throughput, and energy-efficient computation. One of the most significant areas of development is photonic integrated circuit (PICs), which integrate optical components onto a single chip similar to how transistors are integrated in electronic ICs. These chips can manipulate light for computation, routing, and control. In parallel, neural network optical computation standards. A recent article describes the recent advancements of PICs utilizing Indium Phosphide (InP), *InP-Based Generic Foundry Platform for Photonic Integrated Circuits*. InP-based photonic components are mature and reliable sources for communication systems and have been deployed in the field for some time. The increasing need for bandwidth is demanding densely integrated photonic circuits and InP is the material of choice for monolithic integration of active and passive components. This allows a scalable solution for large channel numbers without requiring additional high-precision assembly techniques. The technology allows freedom in the placement of amplifiers and lasers in the circuit, ultra high performance modulators and wide band photodetectors (C- and L-band). Next to this, InP has a number of additional advantages over other integration platforms: because of

the integration of sources and detectors, full circuits can be tested on-wafer, the current fab capacity and supply is line with the demand for PICs, this makes InP a low-cost solution for these requirements [17].

A second implementation of optical processing will be for raw computing clock speed. A computer's clock rate determines the minimum time between sequential operations or instructions. Through Moore's Law we have seen exponential improvements to clock rates and parallel system architectures, but now it has remained stagnant for over two decades. A proposition written in *All-optical computing with beyond 100-GHz clock rates* shows how an all-optical computer can change this. An all-optical computer can realize linear operations, nonlinear functions, and memory entirely in the optical domain with >100 GHz clock rates [18].

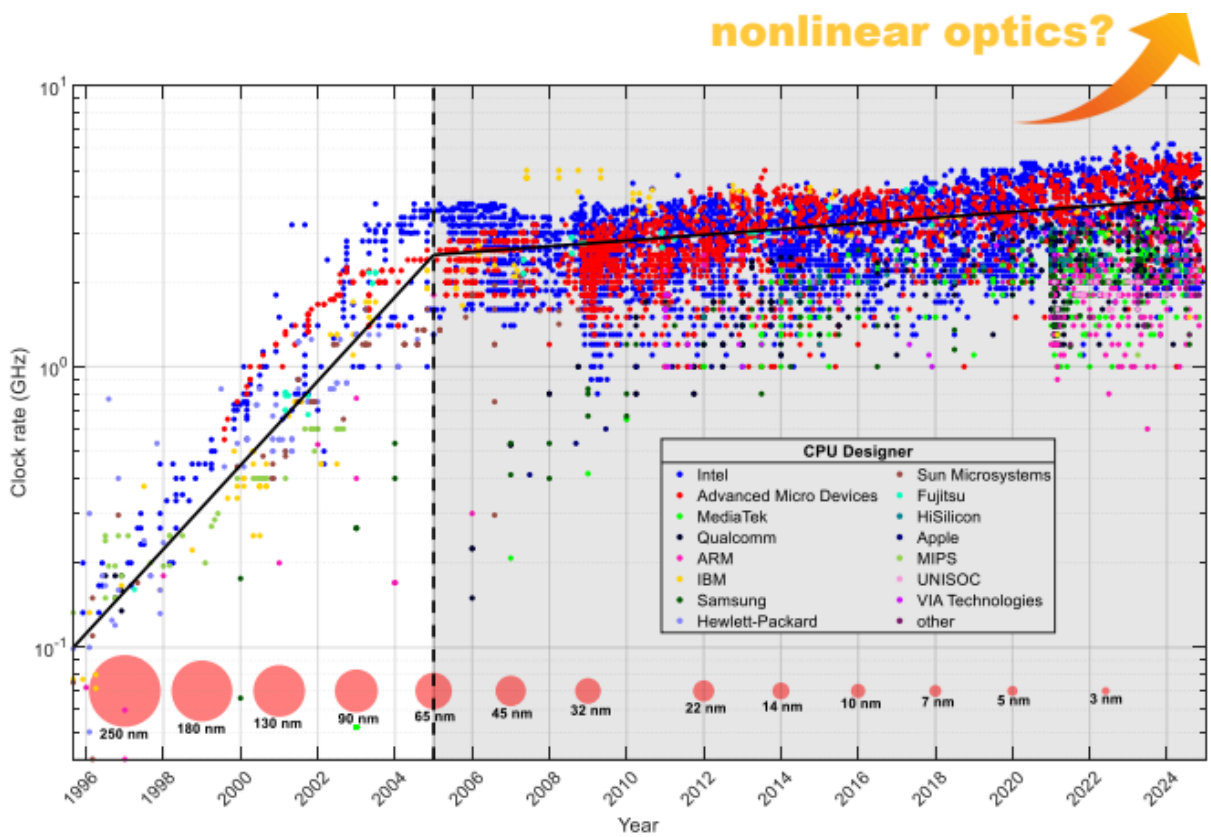


Figure 3.1: CPU Clock rates over the past 29 years. Each point indicates the clock rate and testing date for a different type of commercially-available CPU. The colors represent the entity that designed the CPU, with CPU designers listed. Red circles indicate advanced semiconductor process nodes and around when they were introduced. [18]

Modern CPU clock rates have remained <5 GHz since around 2005. This was realized due to the Von Neumann bottleneck, which refers to a performance limitation in traditional computer architecture where data and instructions are fetched from memory via a single bus. This made CPU designers put their focus not on improving clock rates, but rather increasing performance elsewhere. The addition of more computing cores and advancing computer architectures achieved this.

The paper *All-optical computing with beyond 100-GHz clock rates* discussed how 100 GHz can be achieved using a recurrent neural network (RNN). RNN is well suited for optical computing because it is a driven dynamical system and is well suited for temporal or sequential information processing due to the inherent memory endowed by the recurrent neuron states that are propagated between successive time steps [18].

$$h_i(t+1) = f_i\left(\sum_{j=1}^{N} W_{ij}^{r} \cdot h_j(t) + \sum_{j=1}^{m} W_{ij}^{in} \cdot s_j(t)\right)$$

$$o_k(t) = \sum_{l=1}^{N} W_{kl}^{out} \cdot h_l(t)$$

Figure 3.2: This set of equations represents the core update rule for a Recurrent Neural Network (RNN), more specifically the basic structure of an Echo State Network (ESN). [18]

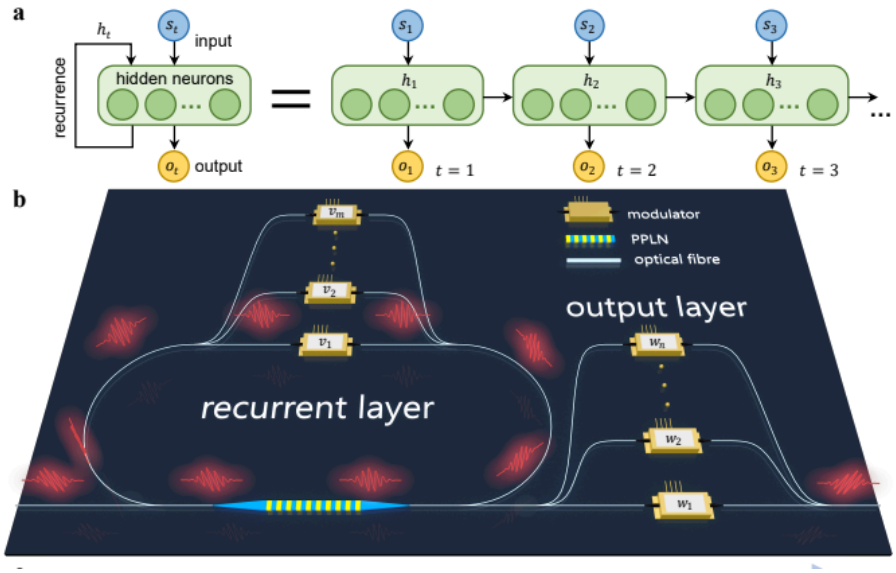


Figure 3.3: All-optical computer architecture (a) A general recurrent neural network (RNN) consists of an input layer, hidden layer with recurrent connections, and output layer. (b) Schematic of the end to end all optical recurrent neural network (AO-RNN) architecture based on a time multiplexed photonic network. [18]

Optical technology presents a compelling pathway to overcome the frequency limitations of traditional electronic CPUs, primarily by leveraging the speed and efficiency of photons. Unlike electrons, photons can travel through waveguides without resistance, eliminating issues such as heat generation, parasitic capacitance, and signal delay that plague electronic circuits. This enables signal propagation at speeds that could support clock frequencies well beyond 100 GHz. Recent advances in silicon photonics have demonstrated the feasibility of integrating optical components onto silicon chips, allowing for ultra-fast data modulation and transmission. Emerging technologies such as optical transistors, Mach-Zehnder interferometers, and nanophotonic logic gates are being developed to perform computation using light, pushing the boundaries of logic operation speed. Additionally, the use of optical clocking via mode-locked lasers enables precise synchronization at extremely high frequencies. While fully optical CPUs remain in the research phase, the industry is moving toward hybrid architectures that combine electronic logic with optical interconnects and co-processors. This gradual integration of photonics into computing platforms represents a realistic path toward achieving and sustaining clock speeds in excess of 100 GHz, with significant gains in performance, parallelism, and energy efficiency.

3.2 - Photonic and Optoelectronic Memory

Innovations to the speed of traditional silicon-based memory have been more diminishing with each new generation of DRAM. To truly regain significant improvements to the data transfer speeds of memory, The Von Neumann bottleneck must be tackled in new ways rather than simply moving DIMMs and cache closer to and/or built into an SoC/Processor. This is where different optical technologies can be integrated into memory in order to gain real ground in innovations and improvements to memory. This section of the paper will go over two different examples of different optical memory technologies.

The first example is magneto-optical memory from a paper *Integrated non-reciprocal magneto-optics with ultra-high endurance for photonic in-memory computing* from Nature Photonics in January of 2025. This work presents a non-reciprocal, non-volatile magneto-optical memory cell designed for high-speed, energy-efficient in-memory computing in the optical domain. The memory cell utilizes a silicon micro-ring resonator (MRR) clad with cerium-substituted yttrium iron garnet (Ce:YIG), exhibiting both magneto-optic (non-reciprocal) and thermo-optic (reciprocal) effects for precise weight programming [19].

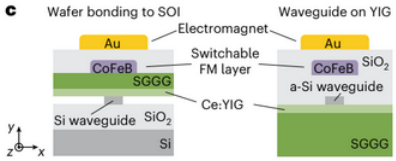


Figure 3.4a

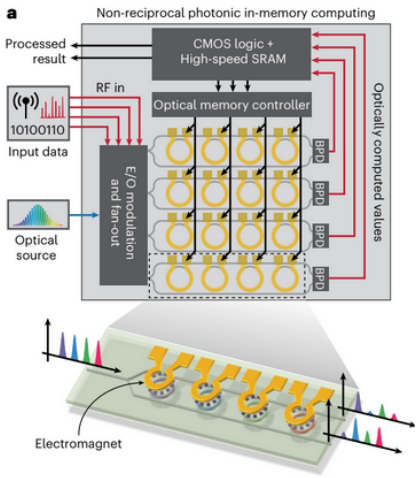


Figure 3.4b

(4a) shows a 2d representation of an electromagnet structure inside of the memory controller. (4b) shows the flow of data through the magneto-optical memory controller as well as the individual electromagnet structures. [19]

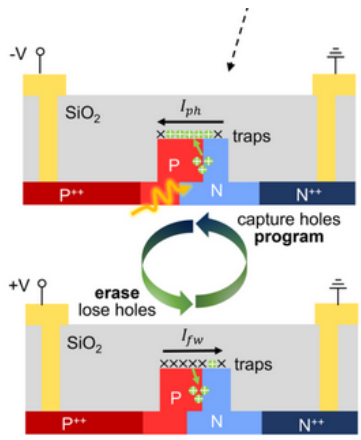
The clockwise (CW) and counter-clockwise (CCW) resonance shifts are decomposed into their thermo-optic (symmetric) and magneto-optic (antisymmetric) components. The thermo-optic effect red-shifts both CW and CCW spectra equally, while the magneto-optic effect introduces a directional shift between them. By analyzing the average shift and split between CW/CCW modes, the team quantifies each component and matches results with a theoretical model. Full width at half maximum (FWHM) linewidths of CW and CCW modes remain nearly identical, indicating similar quality factors. Non-reciprocal loss in Ce:YIG contributes to extinction ratio (ER) variations and linewidth asymmetry. Transmission response for different probe detunings shows how increasing red-detuning enhances the dynamic range of the weight function, albeit at the cost of linearity. At larger detunings and currents, discrepancies arise due to asymmetric FWHM values.

Magneto-optic effects dominate at >1 MHz, allowing sub-nanosecond switching while the thermal response fades. For a 1 ms pulse, both blue (magneto-optic) and red (thermo-optic) shifts appear. For 10 ns pulses, only the fast magneto-optic shift is observed (rise/fall ≈ 0.95 ns), with a ferromagnetic resonance at 0.55 GHz. Eye diagram measurements demonstrate reliable weight updates at 500 Mbps (2.28 pJ) and 1 Gbps (298 fJ). Pulse amplitude modulation level 4 (PAM-4) with ±13.8 mA current achieves four distinct optical weight levels at 143 fJ/bit. Programming speed is currently limited by Ce:YIG's 1 GHz resonance, but faster speeds are possible with alternative materials. Compared to state-of-the-art photonic memory solutions (e.g., phase-change, memristors, ferroelectrics), this magneto-optical approach uniquely offers high-speed, fatigue-free, non-volatile operation ideal for scalable, low-latency optical computing systems [19].

The second example of optical memory is an all-silicon optical memory based on photon avalanche-induced trapping from a paper *All-silicon non-volatile optical memory based on photon avalanche-induced trapping* from Communication Physics in 2025. This paper presents a non-volatile optical analog-trapped memory (ATM) implemented using a standard silicon P-N junction and fabricated with foundry-compatible processes. The device operates by leveraging charge trapping at the Si–SiO_2 interface to modulate the waveguide's (conduit that propagates an

electromagnetic field through a dielectric medium) refractive index, enabling multi-level optical memory storage with zero static power consumption [20].

The ATM utilizes charged interface traps in the SiO_2 layer to adjust the built-in voltage of the P-N junction:



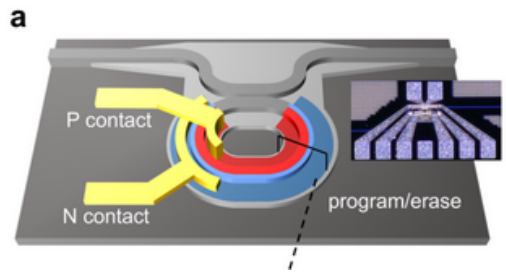


Figure 3.5a

Figure 3.5b

(5a) Shows a 2d model of the PN junction showing the traps and how external light injected causes the photon-induced avalanching holes to be trapped within the oxide layer traps. (5b) Shows a 3D model of the PN Junction ring structure. [20]

Trapped charges modulate the refractive index via the plasma dispersion effect (refractive index of silicon changing when different biases/majority carrier concentrations are present), leading to non-volatile optical resonance shifts. High reverse bias and O-band light (1260-1360 nm) injection initiate avalanche photogeneration, creating a surplus of holes that fill donor-like traps. For erasure, applying a forward bias enables trap discharge, reversing the index change. Once programmed or erased, the state persists without bias due to the hysteresis of trapped charges. The device demonstrates 26 discrete states, repeatable switching behavior over 100 cycles, and stable resonance shifts (~42–71 pm) maintained for over 24 hours. The ATM's non-volatility enables energy-efficient optical neural networks by statically programming weights into MRR(microring-resonator)-based ONNs. On the Modified National Institute of Standards and Technology (which is a database for image processing), the ATM-based ONN achieves 94.8% accuracy using 26 levels (σ ~0.004) [20].

The ATMs are implemented on 220 nm silicon-on-insulator wafers at American Machine and Foundry (AMF) and replicated at CEA-Leti, showing foundry-agnostic compatibility. Resonant wavelength shift is scalable (42–71 pm) via trap density tuning, with extinction ratios (ER) up to 26 dB. The memory enables resonance correction in MRRs for dense wavelength- division multiplexing (DWDM) systems, reducing crosstalk by ~2.5 dB. This mechanism is applicable beyond MRRs to other PIC components, improving latency, integration, and energy efficiency. The switching speed is currently modest due to slow trap charging, which can be improved by engineering trap capture cross-sections, introducing vertical electric fields (faster than flash memory mechanisms), and using shorter-wavelength lasers for more efficient carrier multiplication. The gradual charge accumulation allows for a greater number of intermediate states than most competitors, boosting computational resolution. The silicon-based ATM offers a zero-power, multi-level optical memory that integrates seamlessly into standard photonic platforms without altering fabrication processes. It demonstrates strong non-volatility, excellent retention and endurance, high compatibility, versatility, and superior energy efficiency for ONNs and photonic applications. [20].

These two examples demonstrate the vast optimizations and improvements to performance that can be made when integrating optical technology like magneto-optical weight analysis within a memory controller or utilizing photonic avalanching in semiconductor processes. As silicon and optical innovations continue to merge in relation to memory structure and semiconductor fabrication, we will begin to see large leaps of performance similar to those of early silicon-based memory.

3.3 - Optical Switching and Modulation in Data Centers

In order to continue the trend of progress brought on previously by purely-electrical transistor devices, the proposed optical integrations must continually improve upon existing bit rates/bandwidth, rein in the energy consumption of both individual devices and data centers as a whole, and ensure that the server configurations that compose these data centers are scalable to the needs of any consumer. To this end, multiple groups and studies have proposed a number of solutions, ranging from formatting server structures on a micro-mechanical level with optical switches to modifying the manner in which signals are multiplexed/demultiplexed. These

solutions can lend to either a full replacement of current electrical switch-based networking to a transition period into opto-electronic integration, and each should be considered based on the individual needs of the data center in consideration.

The NEPHELE is one such optical data center network that looks to outright replace electrical switching technologies with optical counterparts. Unlike most modern data centers which rely on field-effect transistor-based electrical switches, the NEPHELE only utilizes high-radix (multiple input, multiple output) optical space switches for computation. These switches are comprised of star couplers, which broadcasts optical signals from any of its input ports to all of its output ports [21]. When used in conjunction with wavelength and spectrum selective switches, these optical switches, which would normally broadcast to every output attached to their coupler, would instead quickly switch between multiple wavelengths of light at up to 200 ns speeds to selectively communicate with each necessary output only. Additionally, since the NEPHELE consists only of optical switches, its transmission speeds operate independently of any network protocols or modulation formats that might otherwise slow down a normal, electrical network. Not only does this cut down on the latency in communications, with average clock recovery times of 25.6 μs and even times of 150 ns observed in burst operation, but the lack of electrical switches within the NEPHELE's design means that there is no need for constant conversion from optical to electrical signal, or vice versa, which lends to being more energy efficient. The largest of the switches tested within the NEPHELE, a 192x192 optical circuit switch, consumes only 75W in normal operation, whereas a more typical 4x4 optical packet switch consumes only 50W [22].

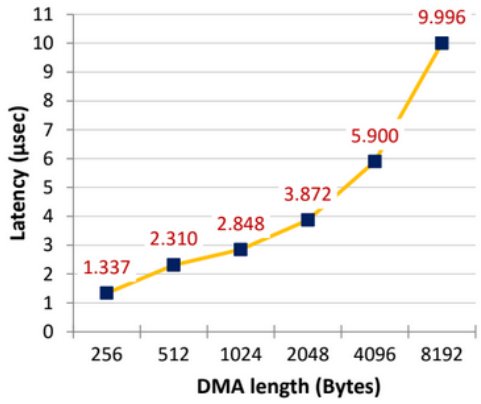


Figure 3.6a

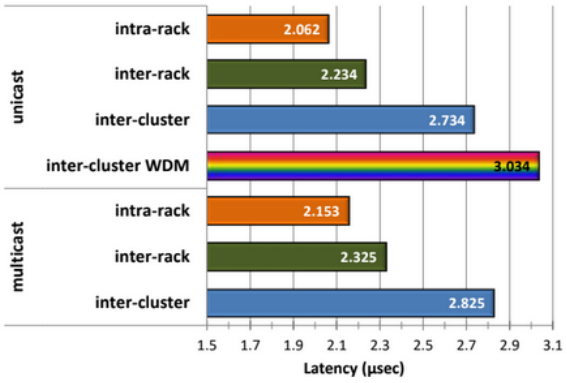


Figure 3.6b

(6a) A measure of direct memory access (DMA) read and write latency vs the length of a packet subject to DMA. (6b) The latency of packet travel between switches in the same rack, in different racks, and between different groups of switches. [22]

In terms of more moderate integration, there has been promise shown in improving upon existing signal conversion technologies. There have been concerns that current analog-to-digital and digital-to-analog converters (ADC/DAC) have been constricting bandwidth when interacting with application-specific integrated circuits. However, designing networks with multiple ADCs and DACs alongside combinations of electronic and optical multiplexing have shown that these constraints can be overcome in multiple different manners. These network configurations utilize two different means of optical and electronic multiplexing in order to increase the number of frequencies that can be accepted as input, combatting part of the difficulties with using modern ADCs and DACs: baseband multiplexing (BB-MUX), which doubles the sampling rate of a signal to extend the upper band of the signal bandwidth before the signal reaches its frequency cutoff, and intermediate-frequency-involved multiplexing (IFI-MUX), which allows for a greater number of frequencies to be allowed by downscaling a signal's strength in order to make it easier to convert between analog or digital [23]. These methods, in conjunction with two different materials of DACs, Complementary Metal-Oxide-Semiconductor (CMOS) and silicon germanium (SiGe), have produced net data transmission speeds of 2.36 Tbps per wavelength between devices, and 2.32 Tbps per wavelength over 80 km single-mode transmission. And since these methods take into account electronic components, these can be more easily integrated into existing systems than outright replacing electronic systems with purely-optical components [24].

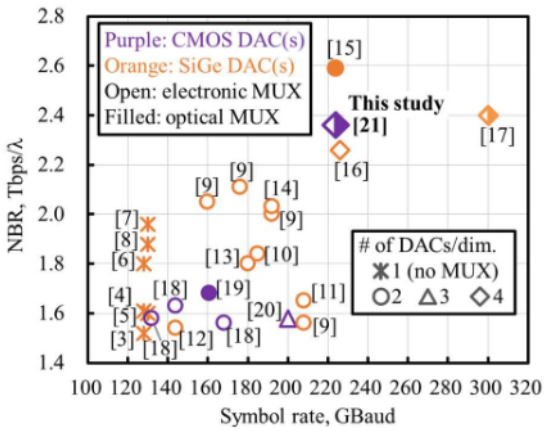


Figure 3.7: A collection of different experiments between DAC material and modalities of multiplexing, organized based on their net bit rate and symbol rate outputs. While there are many different studies listed here, all report speeds above 1 Tbps per wavelength [24].

Summary and Conclusions

With the end of an era of computational development via the miniaturization of existing transistor technology, a dramatic change must be made in order to continue to keep up with the consumer and industry demand for greater personal processing speeds, larger storage capacity, and increased bandwidth allowances for households and businesses. Optical technologies have shown, in the early stages of their research and development, to be able to answer this call to arms, readily taking up the proverbial torch in not only continuing to expand the capabilities of computing devices ever upward but also to improve upon some specific failings that the modern expansion of electrical systems has revealed, such as the cost of power consumption of these systems as they have scaled to larger and larger needs. Integrating optical components into existing electrical technologies would allow us to alleviate some of the growing pains our computing and mass data transit infrastructures have developed in order to keep up with consumer demand, and allow us to continue to expand even further beyond without as steep a cost.

However, the work and research presented herein only scratches the surface of what optical technologies are capable of. As file sizes and the demand to communicate said files increases

with newly-made and newly-expanding connections, so too must the effort to find new ways in which to supply that demand. Further research and development in this field will inevitably yield more ways to collect and process the data of our everyday lives, until such time as either the demand for greater speeds stagnates or we, as a species, find a method of data transmission, or in which a medium in general, surpasses the speed of light, both eventualities being one humans will not have to worry about until many, many years from now.

References

[1] P. Ambs, “Optical Computing: A 60-Year Adventure.” *Advances in Optical Technologies*, vol. 2010, 11 May 2010, pp. 1–15, https://doi.org/10.1155/2010/372652.

[2] S. Dufresne, “Inventing the Microprocessor: The Intel 4004.” *Hackaday*, 30 Jan. 2018, hackaday.com/2018/01/29/inventing-the-microprocessor-the-intel-4004/.

[3] R. Nair, “Evolution of Memory Architecture.” *Proceedings of the IEEE*, vol. 103, no. 8, 1 Aug. 2015, pp. 1331–1345, ieeexplore.ieee.org/abstract/document/7151782, https://doi.org/10.1109/JPROC.2015.2435018.

[4] J. Kim *et al.*, “GRIM-Filter: Fast Seed Location Filtering in DNA Read Mapping Using Processing-in-Memory Technologies" (May 2018). *BMC Genomics.* PubMed Central.

[5] A. Freedman, *The Computer Desktop Encyclopedia.* New York, Amacom, 1999.

[6] R.Balodis, I. Opmane, and A. Tatnall, “History of Data Centre Development”. *Reflections on the History of Computing : Preserving Memories and Sharing Stories*, pp.180-203, 2012, IFIP Advances in Information and Communication Technology (SURVEY), https://doi.org/10.1007/978-3-642-33899-1_13.

[7] J.L.McKenney and A.W. Fisher, “Manufacturing the ERMA Banking System: Lessons from History.” *IEEE Annals of the History of Computing*, vol. 15, no. 4, 1993, pp. 7–26, ieeexplore.ieee.org/abstract/document/238388, https://doi.org/10.1109/85.238388.

[8] J. Daintith, and E. Wright. *A Dictionary of Computing (6 Ed.)*. Oxford University Press, USA, 2008. https://doi.org/10.1093/acref/9780199234004.001.0001

[9] R. Bhargava and K. Troester, *AMD next Generation “Zen 4” Core and 4th Gen AMD EPYC™ 9004 Server CPU*. 27 Aug. 2023, https://doi.org/10.1109/hcs59251.2023.10254726.

[10] J. Choquette, “NVIDIA Hopper H100 GPU: Scaling Performance,” *IEEE Micro*, pp. 1–13, 2023, doi: https://doi.org/10.1109/mm.2023.3256796

[11] J. Macri, “AMD’s Next Generation GPU and High Bandwidth Memory Architecture: FURY,” *Hot Chips GPU Session.* Aug. 2015, https://ieeexplore.ieee.org/abstract/document/7477461, https://doi.org/10.1109/HOTCHIPS.2015.7477461

[12] A. Shilov and P. Alcorn, “Micron shows massive 256GB DDR5-8800 memory sticks — High-capacity double-height 20-watt MCRDIMM modules come in different flavors,” *Tom’s*

Hardware, Mar. 22, 2024.
https://www.tomshardware.com/pc-components/ddr5/micron-shows-massive-256gb-ddr5-8800-memory-sticks-high-capacity-20-watt-mcrdimm-modules-for-next-generation-servers-come-in-different-flavors.

[13] A. Singh *et al.*, "Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network," *COMMUNICATIONS OF THE ACM*, vol. 59, no. 9, 2016, https://dl.acm.org/doi/10.1145/2975159, doi: https://doi.org/10.1145/2975159.

[14] "DATA CENTER NETWORKING QUICK REFERENCE GUIDE," Jan. 2025. Available: https://www.delltechnologies.com/asset/en-us/products/networking/briefs-summaries/Dell_EMC_Networking_-_QRG_-_Data_Center.pdf.

[15] L. A. Barroso, U. Hölzle, and R. Parthasarathy, *The datacenter as a computer : designing warehouse-scale machines*. San Rafael, California: Morgan & Claypool Publishers, 2019.

[16] T. N. Theis and H.-S. P. Wong, "The End of Moore's Law: A New Beginning for Information Technology", *Computing in Science and Engineering,* vol. 19, no. 2, Mar.-Apr. 2017, pp. 41-50, https://ieeexplore.ieee.org/document/7878935, doi: https://doi.org/10.1109/MCSE.2017.29.

[17] L. M. Augustin *et al.*, "InP-Based Generic Foundry Platform for Photonic Integrated Circuits," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 24, no. 1, pp. 1–10, Jan. 2018, doi: https://doi.org/10.1109/jstqe.2017.2720967

[18] L. Gordon *et al.*, "All-optical computing with beyond 100-GHz clock rates," *arXiv.org*, 2025. https://arxiv.org/abs/2501.05756.

[19] P. Pintus *et al.*, "Integrated non-reciprocal magneto-optics with ultra-high endurance for photonic in-memory computing," *Nature Photonics*, vol. 19, Oct. 2024, doi: https://doi.org/10.1038/s41566-024-01549-1

[20] Y. Yuan *et al.*, "All-silicon non-volatile optical memory based on photon avalanche-induced trapping," *Communications Physics*, vol. 8, no. 1, Jan. 2025, doi: https://doi.org/10.1038/s42005-025-01934-4

[21] D. Alistarh *et al.*, "A High-Radix, Low-Latency Optical Switch for Data Centers," *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication*, pp. 367–368, Aug. 2015, doi: https://doi.org/10.1145/2785956.2790035. https://conferences.sigcomm.org/sigcomm/2015/pdf/papers/p367.pdf.

[22] G. M. Saridis *et al.*, “Lightness: A Function-Virtualizable Software Defined Data Center Network With All-Optical Circuit/Packet Switching,” *Journal of Lightwave Technology*, vol. 34, no. 7, pp. 1618–1627, Apr. 2016, doi: https://doi.org/10.1109/jlt.2015.2509476

[23] H. Yamazaki *et al.*, “Analog Electronic and Optical Multiplexing Techniques for Transmitter Bandwidth Extension,” *Journal of Lightwave Technology*, vol. 43, no. 4, pp. 1550–1564, Feb. 2025, https://opg.optica.org/jlt/abstract.cfm?URI=jlt-43-4-1550.

[24] H. Yamazaki *et al.*, “Electro-Optic Bandwidth Quadrupler for Beyond-2-Tbps Transmission,” *Journal of Lightwave Technology*, pp. 1–8, 2025, doi: https://doi.org/10.1109/jlt.2025.3546949